

Litter Physics

A formal model, numerical method,
and implementation audit

Litter Physics project

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Scientific source baseline: 9a66a55

Abstract

This note derives the models currently implemented for a two-box pine-pellet litter simulation, connects their equations to Rust and Python components, and separates numerical evidence from intended physical capability. The household model combines rigid multisphere pellets, frictional contact, timed visit and maintenance events, and conservative but phenomenological water/fines transport. Research fixtures use quadratic MLS-MPM with affine particle velocities, finite-strain viscoplastic paste or weakly compressible liquid, and optional single-pellet impulse coupling. Worked dimensional examples, constitutive calculations, method diagrams and archived refinement plots make assumptions and failure modes explicit. The six-case slump study completes but does not meet its refinement criterion. Neither full physical energy closure nor measured household accuracy is established. No calibrated response bridge between the two numerical scales is yet active.

Status of this document. A source-grounded description of an executable preliminary model, not a claim that the intended research multiphysics or an actual household has been validated. The native fidelity labels remain `preliminary_household` and `research_unvalidated`.

Contents

1	Problem definition, scope, and two numerical scales	3
1.1	Evidence and approximation vocabulary	3
1.2	Notation and units	4
2	Rigid pellets, contact, and explicit sifting geometry	4
2.1	Mass properties and multisphere approximation	4
2.2	Contact law and history	5
2.3	Equations of motion, timestep, and a dimensional check	6
3	Stateful household material and event model	6
3.1	Uptake, damage, swelling, and drainage	6
3.2	Visits, sources, and compliant paw proxies	7
4	Constitutive models implemented in the research solver	8
4.1	Elastic constants, stresses, and shear-rheometer conventions	8
4.2	Trial Hencky strain and the scalar implicit return	8

4.3	Reconstructing the elastic tensor and estimating plastic loss	9
4.4	Worked Herschel-Bulkley return	10
4.5	Weakly compressible liquid: EOS, viscosity, and volume update	10
4.6	Hydrostatic initialization and artificial-compressibility scales	11
5	Research continuum balances and MLS-MPM discretization	12
5.1	Particle and grid variables	12
5.2	Particle-to-grid transfer and explicit stress force	13
5.3	Wall treatment, transfer back, and local update	13
5.4	Adaptive integration and transactional state	13
6	Single-pellet coupling and energy accounting	14
6.1	Impulse exchange, not a porous-mixture closure	14
6.2	Species and linear momentum residuals	14
6.3	Why a stationary wall's projection loss is not external work	15
7	Verification observables and actual refinement evidence	15
7.1	What the fixture observables measure	15
7.2	Separate spatial and temporal experiments	16
8	Calibration, reduced responses, and uncertainty	18
8.1	What can currently be fitted	18
8.2	Intended research-to-household response interface	18
9	Computational contract, reproducibility, and acceptance gaps	19
9.1	Execution and restart semantics	19
9.2	Scaling calculation, not an extrapolated benchmark	19
9.3	Explicitly unresolved physics and numerical acceptance	20
A	Equation-to-code navigation	21
B	Rebuilding the document and reproducing the experiment	22

1 Problem definition, scope, and two numerical scales

The physical target is a household using two stainless-steel Sorstrem sifting boxes filled with pine pellets. Visits disturb and redistribute the bed; liquid and waste are deposited; wet pellets swell and break down; fines sift into a drawer; scooping, drawer emptying and topping up alter the next visit’s initial condition. The mathematical target is therefore a persistent state trajectory, not a collection of independent clean-box scenes:

$$\mathcal{X}(t + \Delta t) = \mathcal{S}_{\Delta t}[\mathcal{X}(t), \boldsymbol{\theta}, \mathcal{E}_{[t, t+\Delta t]}], \quad \mathbf{y}(t) = \mathcal{H}[\mathcal{X}(t)]. \quad (1.1)$$

$\mathcal{X}$ contains pellet poses, velocities and contact histories; material fields and compartments; and cumulative bookkeeping. $\boldsymbol{\theta}$ denotes geometry, material and proxy-behavior parameters. $\mathcal{E}$ is the timed event sequence; $\mathcal{H}$ supplies observables. Checkpointing must preserve this state, not merely a visualization frame.

Two computational scales address different questions. The household scale retains many discrete pellets and long-lived maintenance state using reduced material laws. Research fixtures isolate continuum constitutive behavior and pellet/material interaction at much smaller sizes. The intended eventual reduction is a measured, verified response $R(\boldsymbol{\theta}, \mathbf{z})$ conditioning household rules on local state $\mathbf{z}$. That coupling remains planned, even though a standalone table interpolator already exists.

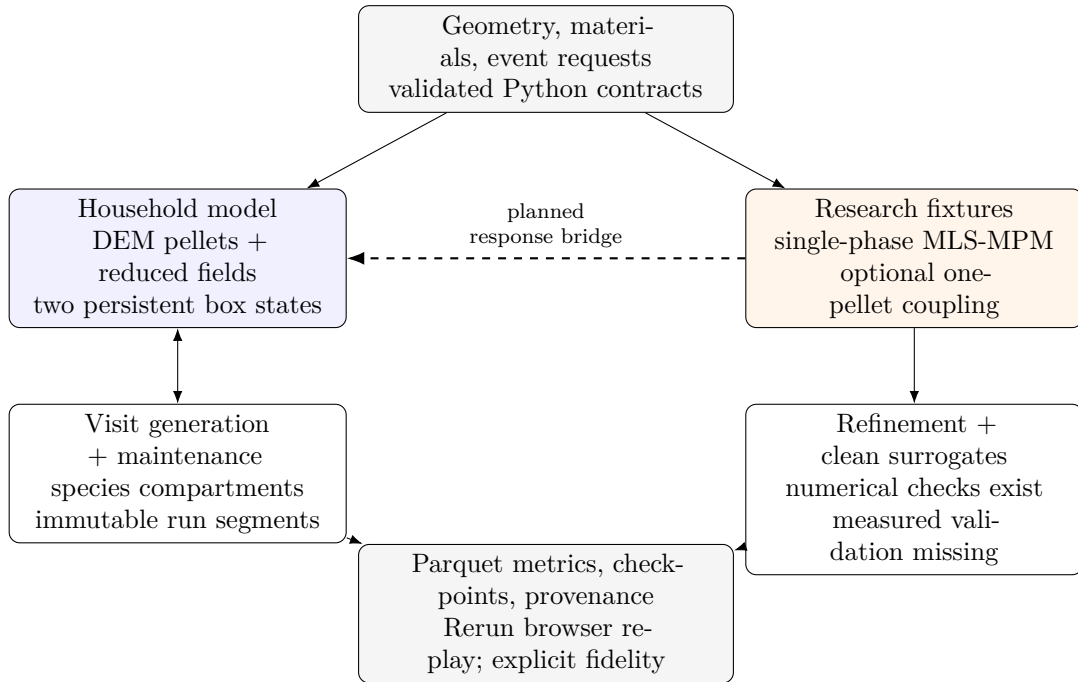


Figure 1: Implemented components use solid arrows; the cross-scale response bridge is not connected. Output and provenance are shared infrastructure, not proof of equivalent fidelity between the two models.

1.1 Evidence and approximation vocabulary

Implemented. A specific update or diagnostic exists in the pinned source. This does not imply accuracy for all inputs.

Verified in a limited sense. A stated numerical property has test or run evidence, with its fixture, tolerance and scope named.

Assumed. A geometry, constant, constitutive closure or behavior distribution lacks the measurements needed to identify it for the intended household.

Planned. A mathematically motivated extension not executed by this version.

The example dimensions and masses are synthetic. No real cat identities, per-cat video measurements, pellet distributions or clean-surrogate material measurements are supplied. A steel-box product name does not determine actual slot geometry, tolerances, packing fraction, friction or wet behavior.

1.2 Notation and units

All solver quantities use SI units; plots sometimes display mm, ms or μJ . Bold lower-case symbols are vectors, bold upper-case symbols are matrices, and $A : B = \text{tr}(A^T B)$ is the Frobenius inner product. $\text{dev } A = A - \frac{1}{3} \text{tr}(A) \mathbf{I}$.

Symbol	Meaning	Units
$h, \Delta t$	Research grid spacing and integration step	m, s
m_p, V_{0p}, J	Material-point mass, reference volume, volume ratio	kg, m^3 , 1
ρ_0, ρ	Reference and current density	kg/m^3
$\mathbf{C}, \mathbf{F}_e$	Affine velocity gradient; elastic deformation	s^{-1} , 1
$\boldsymbol{\sigma}, \boldsymbol{\tau}$	Cauchy and Kirchhoff stresses	Pa
G, K_b, E, ν_P	Shear, bulk, Young's moduli; Poisson ratio	Pa, Pa, Pa, 1
τ_y, K_s, n	Rheometer yield stress, consistency, flow index	Pa, Pa s^n , 1
σ_y, K_e	Equivalent-stress yield and consistency	Pa, Pa s^n
μ, η, ν	Contact friction; dynamic and kinematic viscosity	1, Pa s, m^2/s

The household gravity magnitude is 9.80665 m s^{-2} ; the research wire fixtures use 9.81 m s^{-2} . Worked examples below use the corresponding value. ν_P is Poisson ratio, distinct from kinematic viscosity ν .

2 Rigid pellets, contact, and explicit sifting geometry

The household kernel is a discrete element model (DEM), in the spirit of history-dependent granular contact models [1]. Its pellets are rigid multisphere collision proxies, not deformable wood cylinders. A single pellet has mass m , centre $\mathbf{x}$, orientation $\mathbf{R} \in SO(3)$, velocity $\mathbf{v}$ and angular velocity $\boldsymbol{\omega}$.

2.1 Mass properties and multisphere approximation

For a measured cylinder radius r , length L , and dry density ρ_w , the assigned mass and body-axis moments are

$$m = \rho_w \pi r^2 L, \quad I_{\parallel} = \frac{1}{2} m r^2, \quad I_{\perp} = \frac{1}{12} m (3r^2 + L^2). \quad (2.1)$$

These are *cylinder* properties; summing the overlapping collision spheres would overcount material. Let $a = \max(L - 2r, 0)$. When $a > 0$, set $n_s = \lceil a/r \rceil + 1$ and place sphere centres at

$$s_k = -\frac{a}{2} + \frac{ka}{n_s - 1}, \quad \mathbf{x}_k = \mathbf{x} + \mathbf{R} \mathbf{e}_{\parallel} s_k, \quad k = 0, \dots, n_s - 1. \quad (2.2)$$

For $a = 0$ there is one sphere. Household pellets allow at most 16 spheres. The household body axis is $\mathbf{e}_z$; the separate research pellet uses $\mathbf{e}_x$. Thus their inertia arrays and orientations cannot be copied between implementations without a basis change.

For a slot along x with width w along y , a horizontal pellet at angle θ to the slot has collision-envelope width

$$b_y(\theta) = 2r + a|\sin \theta|. \quad (2.3)$$

The condition $b_y \leq w$ is necessary for this flat orientation to fit between parallel sides; finite slot length, end bars, vertical tilt and other pellets also matter. It is not a stochastic “sift probability.” With the illustrative $r = 3\text{ mm}$ and $w = 5\text{ mm}$, even $b_y(0) = 6\text{ mm} > w$: these *intact* dry pellets cannot pass an ideal slot. Fines are a separate field representation, not smaller DEM spheres generated by fracture.

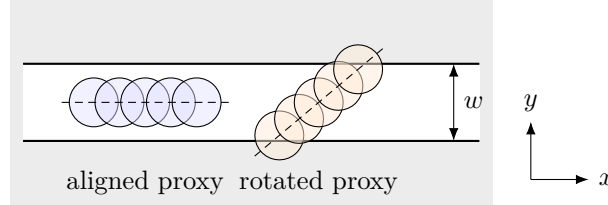


Figure 2: Top-view collision schematic, not measured Sorstrem geometry. Sphere unions set contact geometry while cylinder formulas set mass and inertia. Rotating the same proxy changes its projected width. The drawing uses a slot wider than the pellet diameter for illustration, unlike the synthetic example’s 5 mm slot.

2.2 Contact law and history

Take $\mathbf{n}$ from the second contact partner toward the first. At contact point $\mathbf{x}_c$, the rigid velocity is $\mathbf{u} = \mathbf{v} + \boldsymbol{\omega} \times (\mathbf{x}_c - \mathbf{x})$. For overlap $\delta > 0$, relative velocity $\mathbf{u}_{ab}$, and $v_n = \mathbf{u}_{ab} \cdot \mathbf{n}$, the nonadhesive normal force is

$$f_n = \max(0, k_n \delta - c_n v_n), \quad c_n = 2\zeta \sqrt{k_n m_*}, \quad \zeta = \frac{-\ln e}{\sqrt{\pi^2 + (\ln e)^2}}. \quad (2.4)$$

Here $m_* = m_a m_b / (m_a + m_b)$ for pellet pairs and $m_* = m_a$ for a wall (the driven-tool branch also uses pellet mass). The zero-restitution endpoint uses critical damping, $\zeta = 1$. The restitution conversion is for a linear dashpot idealization; force clipping, multiple sphere contacts and finite steps mean it does not prescribe exact restitution of a whole pellet bed.

Project the previous tangential displacement into the new tangent plane, then advance it with $\mathbf{v}_t = \mathbf{u}_{ab} - v_n \mathbf{n}$:

$$\boldsymbol{\xi}^{\text{tr}} = (\mathbf{I} - \mathbf{n}\mathbf{n}^T) \boldsymbol{\xi}^n + \Delta t \mathbf{v}_t, \quad (2.5)$$

$$\mathbf{f}_t^{\text{tr}} = -k_t \boldsymbol{\xi}^{\text{tr}} - c_t \mathbf{v}_t, \quad k_t = \frac{2}{7} k_n, \quad c_t = \sqrt{\frac{2}{7}} c_n, \quad (2.6)$$

$$\mathbf{f}_t = \mathbf{f}_t^{\text{tr}} \min \left(1, \frac{\mu f_n}{\|\mathbf{f}_t^{\text{tr}}\|} \right). \quad (2.7)$$

Zero trial force is handled without division. When sliding clips the force, the implementation resets $\boldsymbol{\xi} = -\mathbf{f}_t/k_t$; otherwise it retains $\boldsymbol{\xi}^{\text{tr}}$. In particular, this reset does *not* add the $-c_t \mathbf{v}_t/k_t$ correction of some alternative tangential-spring schemes. The rolling torque is

$$\mathbf{T}_r = -\mu_r r_* f_n \frac{\boldsymbol{\omega}_a - \boldsymbol{\omega}_b}{\|\boldsymbol{\omega}_a - \boldsymbol{\omega}_b\|}, \quad (2.8)$$

with zero torque for negligible relative rotation and harmonic effective radius $r_* = r_a r_b / (r_a + r_b)$ for two spheres. It acts on the full relative rotation, not a separately resolved rolling/twisting contact law. Every sphere contact contributes a force and a moment about the body centre. A spatial hash finds candidate pairs; deterministic accumulation rebuilds active contact history.

2.3 Equations of motion, timestep, and a dimensional check

The continuous rigid-body equations represented by the integrator are

$$m\dot{\mathbf{v}} = m\mathbf{g} + \sum_c \mathbf{f}_c, \quad \dot{\mathbf{x}} = \mathbf{v}, \quad (2.9)$$

$$\mathbf{I}_b \dot{\boldsymbol{\omega}}_b = \mathbf{T}_b - \boldsymbol{\omega}_b \times (\mathbf{I}_b \boldsymbol{\omega}_b). \quad (2.10)$$

Velocity is updated before position (symplectic Euler); body-frame angular velocity includes the gyroscopic term. An incremental quaternion rotation is left-multiplied into the orientation and renormalized. The household stability heuristic is

$$\Delta t_{\text{DEM}} = 0.1 \sqrt{m_{\min}/k_n}. \quad (2.11)$$

It is a heuristic, not a proof for an arbitrarily connected multisphere packing. The requested household step is subdivided according to this limit. Overlap, stiffness refinement, timestep refinement and bed settling still need assessment.

For the synthetic example's $r = 3 \text{ mm}$, $L = 12 \text{ mm}$, $\rho_w = 1100 \text{ kg m}^{-3}$, and $k_n = 1000 \text{ N m}^{-1}$,

$$m = 0.373221 \text{ g}, \quad I_{\parallel} = 1.6795 \times 10^{-9} \text{ kg m}^2, \quad I_{\perp} = 5.3184 \times 10^{-9} \text{ kg m}^2, \quad (2.12)$$

$$\Delta t_{\text{DEM}} = 61.0918 \text{ }\mu\text{s}, \quad \delta_{\text{wall}} = \frac{mg}{k_n} = 3.66005 \text{ }\mu\text{m}. \quad (2.13)$$

Thus a 1 ms household step needs 17 substeps for these dry pellets. The static single-contact δ_{wall} is a dimensional check, not the maximum overlap expected in an unrelaxed or driven bed.

Implementation mapping. `core/src/dem.rs`: `Pellet::fit_cylinder`, `World::contact_force`, `Pellet::integrate`, `World::stable_dt`, and `BoxGeometry::plate_contact`. The plate resolves slot edges and bars rather than deleting contacts according to a global open-area fraction. The DEM tests cover bounce, sliding, slot orientation, contact history and tool reactions; those isolated tests do not validate the household scenario.

3 Stateful household material and event model

The two boxes retain material and history across visits. Bed, drawer, floor, removed and evaporated compartments each carry species totals. Coarse field cells carry free water, waste solids, wood fines and a pellet-occupancy proxy. This field is a conservative bookkeeping and transport model, *not* the research MPM continuum described below.

3.1 Uptake, damage, swelling, and drainage

For dry pellet mass m_d , water capacity $W_* = c_w m_d$, uptake rate k_u , and available cell water W_c , the discrete uptake is

$$\Delta W = \min\{W_c, (W_* - W)_+[1 - e^{-k_u \Delta t}]\}, \quad W \leftarrow W + \Delta W. \quad (3.1)$$

The same quantity leaves the field. Define saturation $S = \min(W/W_*, 1)$ and a damage history d :

$$d^{n+1} = d^n + k_b S \chi \Delta t, \quad \chi = \begin{cases} 2, & \|\mathbf{v}\| > 0.01 \text{ m s}^{-1}, \\ 1, & \text{otherwise.} \end{cases} \quad (3.2)$$

At $d \geq 1$, the whole pellet becomes field fines and free water in the bed, or drawer material if below the plate. This deterministic threshold is not a resolved fracture mechanism or a measured

particle-size distribution. Evaporation removes $W[1 - e^{-k_e \Delta t}]$ and credits the evaporated ledger. Uptake, damage, evaporation and transport are sequential, so their ordering is part of the numerical approximation.

The swelling implementation uses

$$s = \left(1 + \frac{W \rho_w}{m_d \rho_{\text{water}}}\right)^{1/3}, \quad m = m_d + W, \quad r = sr_0, \quad \mathbf{I}_b = \frac{m}{m_d} s^2 \mathbf{I}_{b0}. \quad (3.3)$$

The sphere-centre offsets are not scaled. Therefore this is neither an exact isotropically swollen cylinder nor a collision geometry whose union volume equals the added water volume. Occupancy is deposited from dry reference volume and clipped at 0.64. It is not a resolved, moisture-dependent porosity field. These choices preserve accounted species mass but do not establish correct wet packing, inertia or capillary transport. Incoming-water momentum and its associated energy are not resolved in this mass update; the household conservation claim here is species bookkeeping, not a coupled momentum or energy balance.

For cubic field spacing h_f and occupancy o_i , cell water capacity is $C_i = \rho_{\text{water}}(1 - o_i)_+ h_f^3$. Above the plate, downward transfer is

$$\Delta W_{i \rightarrow j} = \min \left\{ (W_i - 0.05 C_i)_+ [1 - e^{-5 \Delta t}], (C_j - W_j)_+ \right\}, \quad (3.4)$$

where the rate 5 has units s^{-1} and j is the cell below i . Bottom-cell outflow uses the cell's sampled slot-open fraction a_i :

$$\Delta W_{i \rightarrow D} = W_i [1 - e^{-5 \Delta t}] a_i (1 - o_i)_+, \quad \Delta M_{i \rightarrow D}^{\text{fines}} = M_i^{\text{fines}} [1 - e^{-0.5 \Delta t}] a_i (1 - o_i)_+. \quad (3.5)$$

Additional first-order settling/lateral redistribution and capacity-limited water overflow conserve the transferred masses. Waste solids do not acquire a solved paste stress, momentum, or spreading law in this model. The numerical rate constants and retention fraction are assumptions, not calibrated Darcy fluxes.

A transparent uptake calculation. With $c_w = 3$ and the dry pellet above, $W_* \simeq 1.11966 \text{ g}$. A dry pellet with unlimited nearby water, $k_u = 0.1 \text{ s}^{-1}$, and a 1 s uptake interval takes up $W_*(1 - e^{-0.1}) \simeq 0.10655 \text{ g}$. This example isolates uptake; a full step can immediately evaporate, move or redistribute some water. At full saturation, $k_b = 0.01 \text{ s}^{-1}$ gives 100 seconds to the damage threshold when undisturbed, or 50 seconds if the speed threshold remains exceeded. These are consequences of the assumed law, not predictions of real pine-pellet breakup.

3.2 Visits, sources, and compliant paw proxies

Python draws visit-stage durations and counts from lognormal variables,

$$\ln T \sim \mathcal{N}(\mu_T, \sigma_T^2), \quad \mathbb{E}[T] = e^{\mu_T + \sigma_T^2/2}. \quad (3.6)$$

The semi-Markov stage path includes approach, enter, inspect, optional digging, posture, elimination, optional covering and exit, with an abandonment branch. Not every label has an independent physical motion or duration. Box preferences, skipping/repeating probabilities, deposit mass and water fraction produce timed wire events. Independent child random streams are spawned per cat in profile order; events are generated before the native simulation, not chosen by a cat reacting to the simulated smell or surface state.

The native paw is a driven sphere of mass $m_t = 0.02 \text{ kg}$. Its target moves along the event direction at 0.15 m s^{-1} . With $k_t^{\text{paw}} = 200 \text{ N m}^{-1}$ and $c_t^{\text{paw}} = 2\sqrt{k_t^{\text{paw}} m_t}$,

$$\mathbf{f}_{\text{drive}} = \text{clip}_{\|\cdot\| \leq 2N} [k_t^{\text{paw}} (\mathbf{x}_{\text{target}} - \mathbf{x}_t) - c_t^{\text{paw}} \mathbf{v}_t], \quad m_t \dot{\mathbf{v}}_t = \mathbf{f}_{\text{drive}} + \mathbf{f}_{\text{contact}}. \quad (3.7)$$

Contact reactions therefore affect the proxy instead of prescribing an infinitely stiff paw trajectory. These fixed native force/motion parameters are not per-cat biomechanics. Scooping, emptying and topping up are explicit material transfers; recording only initial and final boxes would lose the maintenance history needed to interpret them.

Implementation mapping. `core/src/household.rs`: `BoxSim::apply_swelling`, `BoxSim::update_occupancy`, `Simulation::moisture`, `Simulation::apply_events`, `Simulation::mechanics`, `tool_drive` and `Simulation::check_conservation`. `core/src/household/field.rs`: `Field::transport`, `Field::evaporate`, `Field::take_water`. `python/litter_physics/behavior.py`: `LogNormal`, `CatProfile`, `generate_visits` and `_one_visit`.

4 Constitutive models implemented in the research solver

The paste law combines isotropic Hencky elasticity and a principal-frame viscoplastic return, following the general class of Herschel-Bulkley MPM models used for yield-stress materials [4]. The liquid law is a weakly compressible Newtonian approximation. The equations below describe this code, not all features of the cited methods and not measured fecal material.

4.1 Elastic constants, stresses, and shear-rheometer conventions

From Young's modulus E and Poisson ratio ν_P ,

$$G = \frac{E}{2(1 + \nu_P)}, \quad \lambda_L = \frac{E\nu_P}{(1 + \nu_P)(1 - 2\nu_P)}, \quad K_b = \lambda_L + \frac{2G}{3} = \frac{E}{3(1 - 2\nu_P)}. \quad (4.1)$$

The symbol λ_L denotes the first Lamé parameter, not a principal stretch or interpolation weight. The wire contract uses rheometer shear values,

$$\tau_{\text{shear}} = \tau_y + K_s |\dot{\gamma}|^n, \quad (4.2)$$

whereas the internal J2 law uses an equivalent deviatoric stress q and equivalent plastic strain rate $\dot{\epsilon}_p$. In simple shear,

$$q = \sqrt{3} |\tau_{\text{shear}}|, \quad \dot{\epsilon}_p = \frac{|\dot{\gamma}_p|}{\sqrt{3}}, \quad \boxed{\sigma_y = \sqrt{3} \tau_y, \quad K_e = 3^{(1+n)/2} K_s}. \quad (4.3)$$

Thus the internal law is $q = \sigma_y + K_e \dot{\epsilon}_p^n$ during plastic flow. Treating the same numeric K as both K_s and K_e would change the material. Moreover, the code's equivalent stress is formed from *Kirchhoff* stress. Ordinary rheometer shear stress is a Cauchy stress; the identification above presumes $J \simeq 1$. At finite volume change, Cauchy equivalent stress is q/J , so this is not an exact constant-Cauchy-yield law under arbitrary compression.

4.2 Trial Hencky strain and the scalar implicit return

Let $\mathbf{F}_e^n$ be the stored elastic tensor and $\mathbf{C}$ the new affine velocity gradient from the grid. Form

$$\mathbf{F}_e^{\text{tr}} = (\mathbf{I} + \Delta t \mathbf{C}) \mathbf{F}_e^n, \quad (4.4)$$

$$\mathbf{b}^{\text{tr}} = \mathbf{F}_e^{\text{tr}} (\mathbf{F}_e^{\text{tr}})^T = \mathbf{U} \text{diag}(\ell_1^2, \ell_2^2, \ell_3^2) \mathbf{U}^T, \quad (4.5)$$

$$e_i^{\text{tr}} = \ln \ell_i, \quad \bar{e} = \frac{1}{3} \sum_i e_i^{\text{tr}}, \quad \mathbf{e}_d^{\text{tr}} = \mathbf{e}^{\text{tr}} - \bar{e} \mathbf{1}. \quad (4.6)$$

The code diagonalizes $\mathbf{b}^{\text{tr}}$ with a cyclic Jacobi eigensolver; it does not require a separately computed right singular basis. With $\mathbf{s} = \text{dev } \boldsymbol{\tau}$,

$$q_{\text{tr}} = \sqrt{\frac{3}{2} \mathbf{s}_{\text{tr}} : \mathbf{s}_{\text{tr}}} = \sqrt{6} G \|\mathbf{e}_d^{\text{tr}}\|. \quad (4.7)$$

If $q_{\text{tr}} \leq \sigma_y$, the state is elastic. Otherwise the discrete J2 stress reduction and overstress law give

$$q = q_{\text{tr}} - 3G\Delta\varepsilon_p, \quad \Delta\varepsilon_p = \Delta t \left(\frac{q - \sigma_y}{K_e} \right)^{1/n}. \quad (4.8)$$

Eliminating the increment yields the actual scalar residual

$$\boxed{f(q) = q - q_{\text{tr}} + 3G\Delta t \left(\frac{q - \sigma_y}{K_e} \right)^{1/n} = 0}, \quad q \in [\sigma_y, q_{\text{tr}}]. \quad (4.9)$$

For $K_e > 0$, $n > 0$ and $q_{\text{tr}} > \sigma_y$, $f(\sigma_y) < 0$, $f(q_{\text{tr}}) > 0$, and in the bracket interior

$$f'(q) = 1 + \frac{3G\Delta t}{nK_e} \left(\frac{q - \sigma_y}{K_e} \right)^{1/n-1} > 0. \quad (4.10)$$

There is a unique root. The code takes a Newton step only if it is finite and strictly inside the updated bracket; otherwise it biseects. The stopping test uses either residual or bracket width below $10^{-12} \max(q_{\text{tr}}, 1 \text{ Pa})$, with a 200-iteration cap. The monotonicity argument explains robustness, but a finite floating-point iteration can still fail and trigger step rejection. $K_e = 0$ instead uses the perfectly plastic result $q = \sigma_y$ directly.

4.3 Reconstructing the elastic tensor and estimating plastic loss

Set $\alpha = q/q_{\text{tr}}$ in the yielded branch and $\alpha = 1$ in the elastic branch. Then

$$\mathbf{e}^{n+1} = \bar{e}\mathbf{1} + \alpha\mathbf{e}_d^{\text{tr}}, \quad (4.11)$$

$$\mathbf{F}_e^{n+1} = \mathbf{U} \text{diag} \left(\frac{e_1^{n+1}}{\ell_1}, \frac{e_2^{n+1}}{\ell_2}, \frac{e_3^{n+1}}{\ell_3} \right) \mathbf{U}^T \mathbf{F}_e^{\text{tr}}, \quad (4.12)$$

$$\boldsymbol{\tau}^{n+1} = \mathbf{U} \text{diag} \left(2Ge_i^{n+1} + \lambda_L \sum_j e_j^{\text{tr}} \right) \mathbf{U}^T. \quad (4.13)$$

The sum of log strains is unchanged by the return, so plastic correction is isochoric. The stored tensor is the returned *elastic* tensor, not a full accumulated deformation tensor or a separately stored plastic gradient. Inverted/nonfinite trial tensors, stretches below 10^{-6} , eigensolver failure or nonfinite stress reject the trial rather than substituting an undeformed particle. The principal-frame return and the explicit kinematic trial remain discrete constitutive approximations.

The reference-volume Hencky elastic energy used by the diagnostic is

$$\Psi_e = G \sum_i e_i^2 + \frac{\lambda_L}{2} \left(\sum_i e_i \right)^2 = G \|\mathbf{e}_d\|^2 + \frac{K_b}{2} (\text{tr } \mathbf{e})^2. \quad (4.14)$$

The code accumulates the backward-Euler plastic-loss estimate

$$\Delta D_P = V_0 q \Delta\varepsilon_p \geq 0. \quad (4.15)$$

It is useful to distinguish this from the elastic energy released by the *isolated radial-return operation*. Since $\Psi_{e,d} = q^2/(6G)$ and the volumetric part does not change,

$$\Delta \Psi_{e,\text{return}} = \frac{q_{\text{tr}}^2 - q^2}{6G} = \frac{1}{2}(q_{\text{tr}} + q)\Delta\varepsilon_p, \quad (4.16)$$

$$\Delta \Psi_{e,\text{return}} - q\Delta\varepsilon_p = \frac{3}{2}G(\Delta\varepsilon_p)^2. \quad (4.17)$$

This algebraic difference is nonnegative and quadratic in the increment. It is not a bound on the unknown full physical work history during an MPM step, nor does it account for transfers, walls or liquid viscosity. It illustrates why positive plastic loss alone does not close the global discrete energy budget.

4.4 Worked Herschel-Bulkley return

Use the synthetic slump parameters $E = 1000 \text{ Pa}$, $\nu_P = 0.2$, $\tau_y = 30 \text{ Pa}$, $K_s = 10 \text{ Pa s}^{0.6}$ and $n = 0.6$. Equations (4.1) and (4.3) give

$$G = 416.6667 \text{ Pa}, \quad \lambda_L = 277.7778 \text{ Pa}, \quad K_b = 555.5556 \text{ Pa}, \quad (4.18)$$

$$\sigma_y = 51.96152 \text{ Pa}, \quad K_e = 24.08225 \text{ Pa s}^{0.6}. \quad (4.19)$$

For an illustrative trial $q_{\text{tr}} = 80 \text{ Pa}$ and $\Delta t = 1.4 \times 10^{-3} \text{ s}$, the bracket endpoints give $f(\sigma_y) \simeq -28.0385 \text{ Pa}$ and $f(q_{\text{tr}}) \simeq 2.25493 \text{ Pa}$. The scalar solution is

$$q \simeq 78.006 \text{ Pa}, \quad \Delta \varepsilon_p \simeq 0.00159523, \quad (4.20)$$

$$\dot{\varepsilon}_p \simeq 1.139453 \text{ s}^{-1}, \quad \alpha \simeq 0.97507446. \quad (4.21)$$

The unrounded bracketed calculation satisfies the stated residual tolerance. Equation (4.15) gives $\Delta D_P/V_0 = 0.124438 \text{ J m}^{-3}$, whereas the isolated return's elastic-energy decrease is $0.126028 \text{ J m}^{-3}$. Their difference is approximately $0.0015905 \text{ J m}^{-3}$, or 1.28% of the recorded plastic-loss estimate for this local example. These numbers do not represent a complete simulated slump step. The document generator independently bisects the same scalar equation to make this calculation reproducible.

Implementation mapping. `core/src/mpm/constitutive.rs: Material::paste, Material::paste_from_shear_rheology, solve_herschel_bulkley_return, symmetric_eigen3` and `update_paste`. `core/src/mpm/solver.rs` multiplies the returned loss density by reference point volume and accumulates it only for accepted steps.

4.5 Weakly compressible liquid: EOS, viscosity, and volume update

For the liquid wire material `water`, yield stress is zero and $n = 1$; the consistency field acts as dynamic viscosity η . The pair (E, ν_P) is merely a way to specify artificial $K_b = E/[3(1 - 2\nu_P)]$, not a claim that liquid has a static Young's modulus or shear elasticity. Define

$$\rho = \frac{\rho_0}{J}, \quad p = K_b \left(\frac{1}{J} - 1 \right) = K_b \left(\frac{\rho}{\rho_0} - 1 \right), \quad \mathbf{D} = \frac{1}{2}(\mathbf{C} + \mathbf{C}^T). \quad (4.22)$$

The actual discrete update and stress are

$$J^{n+1} = J^n [1 + \Delta t \text{tr } \mathbf{C}], \quad (4.23)$$

$$\boldsymbol{\tau}^{n+1} = -p^{n+1} J^{n+1} \mathbf{I} + 2\eta J^{n+1} \text{dev } \mathbf{D}. \quad (4.24)$$

The tensor-history slot is reset to identity for liquid; J supplies volume history. There is no pressure projection, cavitation clamp or pore-pressure mixture coupling. Expanded states $J > 1$ can produce negative EOS pressure. States with $J^{n+1} \leq 10^{-6}$ or nonfinite values are rejected.

Equation (4.23) is a first-order approximation to $\dot{J} = J \text{tr } \nabla \mathbf{v}$, *not* $J^n \det(\mathbf{I} + \Delta t \mathbf{C})$. For $\mathbf{C} = -\mathbf{I} \text{ s}^{-1}$, $J^n = 1$ and $\Delta t = 1 \times 10^{-3} \text{ s}$, these expressions give 0.997000000 and 0.997002999, respectively. The discrepancy is order Δt^2 per step; neither expression is the exact finite-time exponential for constant $\mathbf{C}$. No determinant replacement is silently assumed in this derivation.

The liquid elastic-energy density is

$$\Psi_l(J) = K_b(J - 1 - \ln J), \quad J \frac{d\Psi_l}{dJ} = K_b(J - 1) = -pJ. \quad (4.25)$$

This derivative agrees with the isotropic Kirchhoff stress. Viscous stress is computed, but its dissipated energy is not accumulated in the current global energy diagnostic. The continuum viscous power density would be $2\eta \text{dev } \mathbf{D} : \text{dev } \mathbf{D}$ per current volume; its consistent discrete integral remains a missing ledger term.

4.6 Hydrostatic initialization and artificial-compressibility scales

Let $d = H_0 - z$ denote depth in the placed current configuration. For liquid, $dp/dd = \rho g$ and $p(0) = 0$ give

$$\frac{dp}{dd} = \rho_0 g \left(1 + \frac{p}{K_b}\right) \implies p(d) = K_b \left(e^{\rho_0 g d / K_b} - 1\right), \quad J(d) = e^{-\rho_0 g d / K_b}. \quad (4.26)$$

This is an exact *continuum* hydrostatic solution for the specified EOS. A placed subcell of current volume V is assigned $V_0 = V/J$ and $m = \rho_0 V_0$; assigning $m = \rho_0 V$ instead would discard the compressible density profile. Accurate initial pressure values do not establish discrete force balance or time-evolved hydrostatic accuracy.

Paste initialization instead uses a confined-compression reference, with $M = \lambda_L + 2G$,

$$s = -\frac{\rho_0 g d}{M}, \quad \mathbf{F}_e = \text{diag}(1, 1, e^s), \quad \boldsymbol{\tau} = \text{diag}(\lambda_L s, \lambda_L s, M s). \quad (4.27)$$

It matches the *small-strain* vertical reference $-\tau_{zz} = \rho_0 g d$. At finite strain, density is ρ_0/e^s and Cauchy stress is $\boldsymbol{\tau}/e^s$, so this is not an exact finite-strain hydrostatic solution. It is also a prestressed initializer, not an equilibrium solve subject to the paste yield criterion. For the soft slump values and $H_0 = 20$ mm, $\rho_0 g H_0 / M \simeq 0.177$, not an asymptotically small strain.

The acoustic speeds used by the timestep heuristic are

$$c_{\text{paste}} = \sqrt{\frac{K_b + 4G/3}{\rho_0}}, \quad c_{\text{liquid}} = \sqrt{\frac{K_b}{\rho_0}}, \quad \nu_{\text{liquid}} = \frac{\eta}{\rho_0}. \quad (4.28)$$

Paste has zero explicit diffusivity in this heuristic because its viscous return is local/implicit; global kinematics and stress forces still have an acoustic step restriction. Increasing K_b suppresses artificial compression while increasing acoustic cost. Useful scale checks are

$$\epsilon_c = \frac{\rho_0 g H_0}{K_b}, \quad \text{Ma} = \frac{U}{c}, \quad (4.29)$$

with small ϵ_c and Mach number desirable for a near-incompressible approximation, but neither a substitute for boundary and pressure verification.

For $E = 1 \times 10^5$ Pa, $\nu_P = 0.2$, $\rho_0 = 1000$ kg m⁻³, $\eta = 1 \times 10^{-3}$ Pa s and $H_0 = 10$ mm,

$$K_b = 55\,555.6 \text{ Pa}, \quad c = 7.453\,56 \text{ m s}^{-1}, \quad (4.30)$$

$$p(H_0) = 98.1867 \text{ Pa}, \quad J(H_0) = 0.998\,236. \quad (4.31)$$

The incompressible comparison is $\rho_0 g H_0 = 98.1$ Pa; the artificial compression is about 0.176%. With $h = 2$ mm, CFL = 0.3 and negligible speed, the acoustic limit is $8.049\,84 \times 10^{-5}$ s; the viscous limit $h^2/(4\nu)$ is 1 s. A water-like bulk modulus of 2.2×10^9 Pa at the same reference density would instead give $c \simeq 1483$ m s⁻¹ and an acoustic limit near 4.05×10^{-7} s, about 199 times smaller. This cost tradeoff changes the modeled material; it is not a free numerical acceleration of identical water physics.

Implementation mapping and unresolved accuracy. `core/src/mpm/constitutive.rs`: `Material::liquid`, `update_liquid`, `liquid_pressure`, `wave_speed` and `diffusivity`. `core/src/mpm/fixtures.rs`: `FixtureSpec::hydrostatic_state`, `hydrostatic_reference` and `place_particles`. `core/src/mpm/solver.rs`: `Simulation::elastic_energy`. Existing audits report unresolved bottom-boundary hydrostatic error and timestep-insensitive per-particle liquid pressure noise. Initialization tests, correct EOS algebra and a converged local return do not resolve those spatial/discrete issues.

5 Research continuum balances and MLS-MPM discretization

Research fixtures solve *one* continuum phase (paste or liquid), optionally coupled to one rigid pellet. They do not yet solve a mixture of water, fines and pellets. For current density ρ , velocity $\mathbf{v}$ and Cauchy stress $\boldsymbol{\sigma}$, the local balances motivating the discretization are

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0, \quad \rho \frac{D\mathbf{v}}{Dt} = \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{g}. \quad (5.1)$$

The reference-volume stress is Kirchhoff stress, $\boldsymbol{\tau} = J\boldsymbol{\sigma}$, where J is current volume divided by reference volume. The constitutive history is phase dependent: paste stores an elastic deformation tensor; liquid stores a scalar volume ratio. Neither phase stores the full multiphase state of the intended porous material.

5.1 Particle and grid variables

A material point p stores m_p , V_{0p} , $\mathbf{x}_p$, $\mathbf{v}_p$, an affine velocity matrix $\mathbf{C}_p$, stress and constitutive history. The regular grid has spacing h and positions $\mathbf{x}_i$. Material points carry history across steps; grid mass and momentum are rebuilt each step. The affine representation approximates local velocity as

$$\mathbf{v}(\mathbf{x}) \simeq \mathbf{v}_p + \mathbf{C}_p(\mathbf{x} - \mathbf{x}_p). \quad (5.2)$$

This reduces the loss of rotational/local linear motion characteristic of plain particle-in-cell transfer [2]. It does not guarantee energy conservation.

Let $q = |r|$ for normalized distance $r = (x_i - x_p)/h$. The quadratic B-spline is

$$N(r) = \begin{cases} \frac{3}{4} - r^2, & 0 \leq q < \frac{1}{2}, \\ \frac{1}{2}(\frac{3}{2} - q)^2, & \frac{1}{2} \leq q < \frac{3}{2}, \\ 0, & q \geq \frac{3}{2}. \end{cases} \quad w_{ip} = \prod_{\alpha=1}^3 N\left(\frac{x_{i\alpha} - x_{p\alpha}}{h}\right). \quad (5.3)$$

Each point uses a $3^3 = 27$ node stencil. With complete support,

$$\sum_i w_{ip} = 1, \quad \sum_i w_{ip} \mathbf{d}_{ip} = \mathbf{0}, \quad \sum_i w_{ip} \mathbf{d}_{ip} \mathbf{d}_{ip}^T = \frac{h^2}{4} \mathbf{I}, \quad \mathbf{d}_{ip} = \mathbf{x}_i - \mathbf{x}_p. \quad (5.4)$$

The last identity supplies the $4/h^2$ MLS factor, rather than a separately coded derivative of each shape function. Boundary padding preserves a full stencil for valid particles; leaving that padded grid is not silently accepted as an open boundary in the current closed wire fixtures.

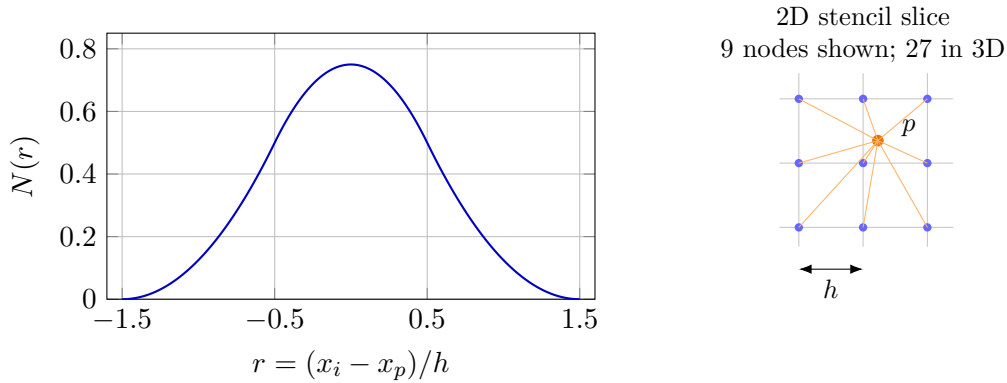


Figure 3: Analytical interpolation kernel and an illustrative stencil slice. These are method diagrams, not simulation output. Exactly zero-weight stencil entries must not create duplicate active grid nodes.

5.2 Particle-to-grid transfer and explicit stress force

The implemented MLS-MPM transfer [3] combines affine momentum and an explicit stress impulse:

$$m_i = \sum_p w_{ip} m_p, \quad (5.5)$$

$$\mathbf{p}_i^* = \sum_p w_{ip} \left[m_p \mathbf{v}_p + \left(m_p \mathbf{C}_p - \Delta t \frac{4}{h^2} V_{0p} \boldsymbol{\tau}_p \right) \mathbf{d}_{ip} \right], \quad (5.6)$$

$$\mathbf{v}_i^* = \frac{\mathbf{p}_i^*}{m_i} + \Delta t \mathbf{g}, \quad m_i > 0. \quad (5.7)$$

In weak-form language the stress contribution is $-\Delta t \sum_p V_{0p} \boldsymbol{\tau}_p \nabla w_{ip}$ with the MLS gradient surrogate $\nabla w_{ip} \simeq (4/h^2) w_{ip} \mathbf{d}_{ip}$. Dimensional check: $V_0 \tau$ has units of energy; multiplying by $\Delta t \mathbf{d}/h^2$ gives momentum. Equation (5.4) cancels internal stress impulses in the sum over nodes. That cancellation can fail in bookkeeping if an active node is processed twice; the zero-deposit regression explicitly guards this case.

5.3 Wall treatment, transfer back, and local update

At a wall with inward unit normal $\mathbf{n}$, let $v_n = \mathbf{v}^* \cdot \mathbf{n}$ and $\mathbf{t} = \mathbf{v}^* - v_n \mathbf{n}$. Only $v_n < 0$ is constrained:

$$\mathbf{v}^N = \mathbf{t}, \quad \mathbf{v}^+ = \mathbf{t} \max \left(0, 1 - \frac{\mu(-v_n)}{\|\mathbf{t}\|} \right), \quad (5.8)$$

with the sticking/zero-tangent case handled explicitly. Axis-aligned wall constraints are applied sequentially at corners. This is a nodal velocity projection with Coulomb tangential reduction. It is *not* an incompressible pressure-Poisson solve and does not enforce a pointwise watertight surface.

After wall and optional pellet coupling updates,

$$\mathbf{v}_p^{n+1} = \sum_i w_{ip} \mathbf{v}_i^+, \quad (5.9)$$

$$\mathbf{C}_p^{n+1} = \frac{4}{h^2} \sum_i w_{ip} \mathbf{v}_i^+ \mathbf{d}_{ip}^T, \quad (5.10)$$

$$\mathbf{x}_p^{n+1} = \mathbf{x}_p^n + \Delta t \mathbf{v}_p^{n+1}. \quad (5.11)$$

Paste uses the trial elastic tensor $\mathbf{F}_e^{\text{tr}} = (\mathbf{I} + \Delta t \mathbf{C}_p^{n+1}) \mathbf{F}_e^n$; liquid advances its scalar J with $\text{tr} \mathbf{C}_p^{n+1}$. The constitutive operations are given in Section 4. A locally implicit plastic return does not make the global momentum step implicit: the stress in (5.6) is the stored stress from the preceding accepted state.

5.4 Adaptive integration and transactional state

The research solver forms a heuristic upper bound

$$\Delta t_{\text{stab}} = \min \left\{ \frac{\text{CFL } h}{c + \max_p \|\mathbf{v}_p\|}, \quad \frac{h^2}{4\nu}, \quad 0.2 \sqrt{m_r/k_n}, \quad \frac{\text{CFL } h}{\|\mathbf{v}_r\|} \right\}, \quad (5.12)$$

where viscous and rigid terms apply only when present/nonzero. Here c is the material wave speed and ν the implementation's kinematic diffusivity. A recovery factor can reduce the candidate step further. Endpoint clipping then lands on the requested record/final time. Nonfinite state, constitutive failure or excessive displacement rejects the trial; the solver restores physical history and ledgers, halves the recovery scale and retries. A minimum step failure is reported instead of silently changing the physics.

The checkpoint persists limiter/rejection history. A timestep-refinement case is not accepted merely because one step happened to reach the requested cap: `limited_steps` counts accepted steps that were limited before endpoint clipping. This distinguishes the tested temporal axis from an adaptive method that actually reused the same smaller steps at every requested resolution.

Implementation mapping. `core/src/mpm/grid.rs`: `GridLayout::stencil`, `Stencil`, `Grid::deposit`. `core/src/mpm/particles.rs`: `ParticleSet`. `core/src/mpm/solver.rs`: `Simulation::particle_to_grid`, `grid_update`, `grid_to_particle`, `try_step`, `stability_limit` and `advance_to`. `core/tests/grid_transfer.rs` tests zero-weight bookkeeping and momentum.

6 Single-pellet coupling and energy accounting

6.1 Impulse exchange, not a porous-mixture closure

For a research pellet, the signed distance is the minimum distance to its multisphere union. A grid node inside that union is constrained only when its relative normal velocity is approaching. With node position $\mathbf{x}_i$,

$$\mathbf{u}_r(\mathbf{x}_i) = \mathbf{v}_r + \boldsymbol{\omega}_r \times (\mathbf{x}_i - \mathbf{x}_r), \quad \mathbf{u}_{\text{rel}} = \mathbf{v}_i - \mathbf{u}_r(\mathbf{x}_i). \quad (6.1)$$

Apply (5.8) to $\mathbf{u}_{\text{rel}}$, then add back $\mathbf{u}_r$. If $\Delta \mathbf{p}_i = m_i(\mathbf{v}'_i - \mathbf{v}_i)$ is the grid impulse, the pellet receives

$$\Delta \mathbf{P}_r = - \sum_i \Delta \mathbf{p}_i, \quad \Delta \mathbf{L}_r = - \sum_i (\mathbf{x}_i - \mathbf{x}_r) \times \Delta \mathbf{p}_i. \quad (6.2)$$

Linear and angular action-reaction balance is explicit at this grid/pellet exchange. The body's start-of-step velocity is used for all constrained nodes, then the summed impulse updates the body. It is not a simultaneous finite-mass contact solve, and the two kinetic-energy changes need not cancel or sum to a nonpositive number. Resolution-dependent leakage, work accuracy and multiple research pellets remain acceptance gaps.

The separate research pellet uses world angular momentum and cylinder inertia; wall contact is spring-dashpot/Coulomb contact rather than the grid projection. The two rigid-body implementations share concepts, not identical state formats or all contact-history details.

6.2 Species and linear momentum residuals

For household species $s \in \{\text{wood, waste solids, water}\}$,

$$r_s = \sum_{c \in \{B, D, F, R, E\}} M_{s,c} - M_{s,0} - M_{s,\text{added}}. \quad (6.3)$$

The compartments are bed, drawer, floor, removed and evaporated, summed over boxes where appropriate. Reclassification of intact wood into fines is not a new source. The native check bounds each residual by $\max(1 \times 10^{-12} \text{ kg}, 10^{-6} M_{s,\text{expected}})$. This tests bookkeeping, not spatial distribution or transport accuracy.

For the research continuum and optional pellet, let $\mathbf{P}$ be the sum of particle translational and pellet linear momentum. The diagnostic is

$$\mathbf{r}_P = \mathbf{P}(t) + \mathbf{P}_{\text{out}}(t) - \mathbf{P}(0) - \mathbf{I}_g - \mathbf{I}_{\text{wall}} - \mathbf{I}_{r,\text{wall}}. \quad (6.4)$$

The internal coupling impulse is not subtracted a second time. Current closed fixtures reject nonzero outflow even when it balances this ledger. Consequently, “mass conserved including escaped material” cannot conceal a wall leak.

6.3 Why a stationary wall's projection loss is not external work

For one normal wall projection, record

$$\Delta E_N = \frac{1}{2}m_i(\|\mathbf{v}^N\|^2 - \|\mathbf{v}^*\|^2) = -\frac{1}{2}m_i v_n^2 \leq 0, \quad \Delta D_F = \frac{1}{2}m_i(\|\mathbf{t}\|^2 - \|\mathbf{v}^+\|^2) \geq 0. \quad (6.5)$$

Thus $\Delta K_{\text{grid}} = \Delta E_N - \Delta D_F$ exactly, up to roundoff. A stationary wall has zero external boundary work, since its boundary velocity is zero. E_N mixes impact-like kinetic loss with numerical operator splitting; calling it wall work is physically misleading.

For example, a resting node given gravity velocity $-g\Delta t$ and then projected back to zero loses $m_i g^2 \Delta t^2 / 2$ on the transient grid each step. Over duration T , this is $m_i g^2 T \Delta t / 2$, which vanishes linearly with timestep. It does not mean the stationary wall physically absorbed that amount of gravitational work from an unmoving particle.

The recorded mechanical energy is

$$E_{\text{mech}} = \sum_p \left[\frac{1}{2}m_p \|\mathbf{v}_p\|^2 - m_p \mathbf{g} \cdot \mathbf{x}_p + V_{0p} \Psi_p \right] + K_r - m_r \mathbf{g} \cdot \mathbf{x}_r. \quad (6.6)$$

Here K_r includes rigid rotation, while the particle kinetic sum does *not* include APIC affine kinetic modes. The accumulated diagnostic sum is

$$\begin{aligned} L_E &= E_N - D_F + W_{r,\text{contact}} + \Delta K_{\text{coupling,grid}} + \Delta K_{\text{coupling,rigid}} - D_P, \\ r_E &= E_{\text{mech}}(t) - E_{\text{mech}}(0) - L_E. \end{aligned} \quad (6.7)$$

$D_P \geq 0$ is the constitutive plastic-loss estimate. $W_{r,\text{contact}}$ is the start-of-step contact-force work estimate; it includes recoverable penalty-spring effects and is not pure dissipation. The coupling terms are measured kinetic changes, not an invented cancellation assumption.

Equation (6.7) is not a closed physical first law. Missing terms include particle/grid transfer and affine energy, liquid viscous loss, complete contact spring/rotation bookkeeping and any outflow energy. Gravity discretization can also contribute a defect. A small r_E alone could reflect cancellation; a nonzero value cannot simply be called conservation success. Future energy acceptance needs a consistently discretized balance, not a renaming of the residual.

Implementation mapping. `core/src/mpm/rigid.rs`: `Pellet::signed_distance`, `couple_grid`, `Pellet::apply_impulse`. `core/src/mpm/solver.rs`: `Ledger`, `Ledger::ledgered_energy`, `Simulation::energy_residual`, `Simulation::momentum_residual`. `core/src/research.rs`: `check_closed_state` and `restore validation`. The restored initial mechanical-energy baseline is checked against a rebuilt initial state; missing histories are rejected rather than fabricated as zero.

7 Verification observables and actual refinement evidence

7.1 What the fixture observables measure

The particle spacing at initialization is $s_p = h/2$. The reported surface proxies are

$$H_h = \max_p z_p + s_p/2, \quad B_{x,h} = \max_p x_p - \min_p x_p + s_p, \quad S_h = H_0 - H_h. \quad (7.1)$$

These estimate height, horizontal extent and slump using particle extrema plus an *initial* spacing correction. They are not reconstructed deformed particle surfaces or a continuum free-surface mesh. They can therefore have quadrature and surface-estimation error in addition to solver error.

The hydrostatic error metrics compare every particle, including those next to the bottom boundary, to the phase-appropriate reference in Section 4:

$$e_{\text{rms}} = \left[\frac{1}{N_p} \sum_p \left(\frac{q_p - q_{\text{ref}}(z_p)}{\rho_0 g H_0} \right)^2 \right]^{1/2}, \quad e_{\text{max}} = \max_p \frac{|q_p - q_{\text{ref}}(z_p)|}{\rho_0 g H_0}. \quad (7.2)$$

For liquid $q = p$ is pressure; for paste it is negative vertical Kirchhoff stress. Removing bottom-row particles or smoothing per-particle liquid pressure would change the acceptance question, not solve the observed accuracy problem. The archived dataset below contains slump results only; it establishes no hydrostatic-pressure convergence claim.

The code also reports a cylinder-slump sanity estimate [5],

$$Y = \frac{\tau_y}{\rho_0 g H_0}, \quad \frac{S_{\text{cyl}}}{H_0} = 1 - 2Y[1 - \ln(2Y)], \quad 0 < Y < \frac{1}{2}. \quad (7.3)$$

It uses zero slump for $Y \geq 1/2$ and H_0 for $Y \leq 0$. For $\tau_y = 30 \text{ Pa}$, $\rho_0 = 1000 \text{ kg m}^{-3}$, $H_0 = 20 \text{ mm}$ and $g = 9.81 \text{ m s}^{-2}$, $Y \simeq 0.152905$ and $S_{\text{cyl}} \simeq 6.63737 \text{ mm}$. The simulated block is a *cube*, with finite elasticity, wall friction and time-dependent rheology. A mismatch with this cylinder estimate is not by itself a validated error measure for that different problem.

7.2 Separate spatial and temporal experiments

For observable Q at three or more successively finer resolutions, the implemented criterion computes

$$\epsilon_k = \frac{|Q_k - Q_{k-1}|}{\max(|Q_k|, Q_{\text{scale}})}, \quad \epsilon_{\text{final}} < 0.05. \quad (7.4)$$

A finest value below Q_{scale} is *uninformative*, not a pass. The threshold is strict. Missing/nonfinite values, incomplete runs, changed fidelity, wrong final times, adaptive limiting, rejected steps and budget overruns prevent passing claims. The spatial axis holds the requested timestep fixed; the temporal axis holds h fixed and requires increasing actual step counts.

Three values can expose a failure to resolve an observable, but a small final change does not prove asymptotic convergence. For equal refinement ratio r ,

$$p_{\text{obs}} = \frac{\ln(|Q_0 - Q_1|/|Q_1 - Q_2|)}{\ln r} \quad (7.5)$$

is merely a trend diagnostic without a verified leading-error expansion. Neither Richardson extrapolation nor a formal discretization uncertainty bound is claimed by this study.

The frozen run uses `examples/research_slump.yaml`, modified only by the study axes in `examples/verification_slump.yaml`: a 20 mm cube in a 60 mm domain, $\rho_0 = 1000 \text{ kg m}^{-3}$, $E = 1000 \text{ Pa}$, $\nu_P = 0.2$, $\tau_y = 30 \text{ Pa}$, $K_s = 10 \text{ Pa s}^{0.6}$, $n = 0.6$, seed 7, final time 0.5 s. All material values are synthetic. The undeformed block mass is 8 g, with continuum initial potential energy $mgH_0/2 = 0.7848 \text{ mJ}$. Non-hydrostatic fixtures add bounded position jitter of amplitude $0.1s_p$; therefore the discrete initial potential energy differs slightly from this integral. Spatial refinement also refines particle quadrature and jitter amplitude; it is not a grid-only experiment with fixed material points.

Axis	h (mm)	Δt (ms)	Points	Steps	Slump (mm)	Spread (mm)
Spatial	10	0.05	64	10000	3.9897	22.6655
Spatial	5	0.05	512	10000	4.3623	23.5512
Spatial	2.5	0.05	4096	10000	4.5165	25.9463
Temporal	5	0.4	512	1250	4.7376	25.0541
Temporal	5	0.2	512	2500	4.6077	24.6164
Temporal	5	0.1	512	5000	4.4875	24.1053

Table 1: Actual compiled-kernel results at 0.5 s, source revision 9a66a55. Every case completed with zero limited and rejected steps. Times and lengths here are requested caps/grid scales and measured final surface proxies, respectively.

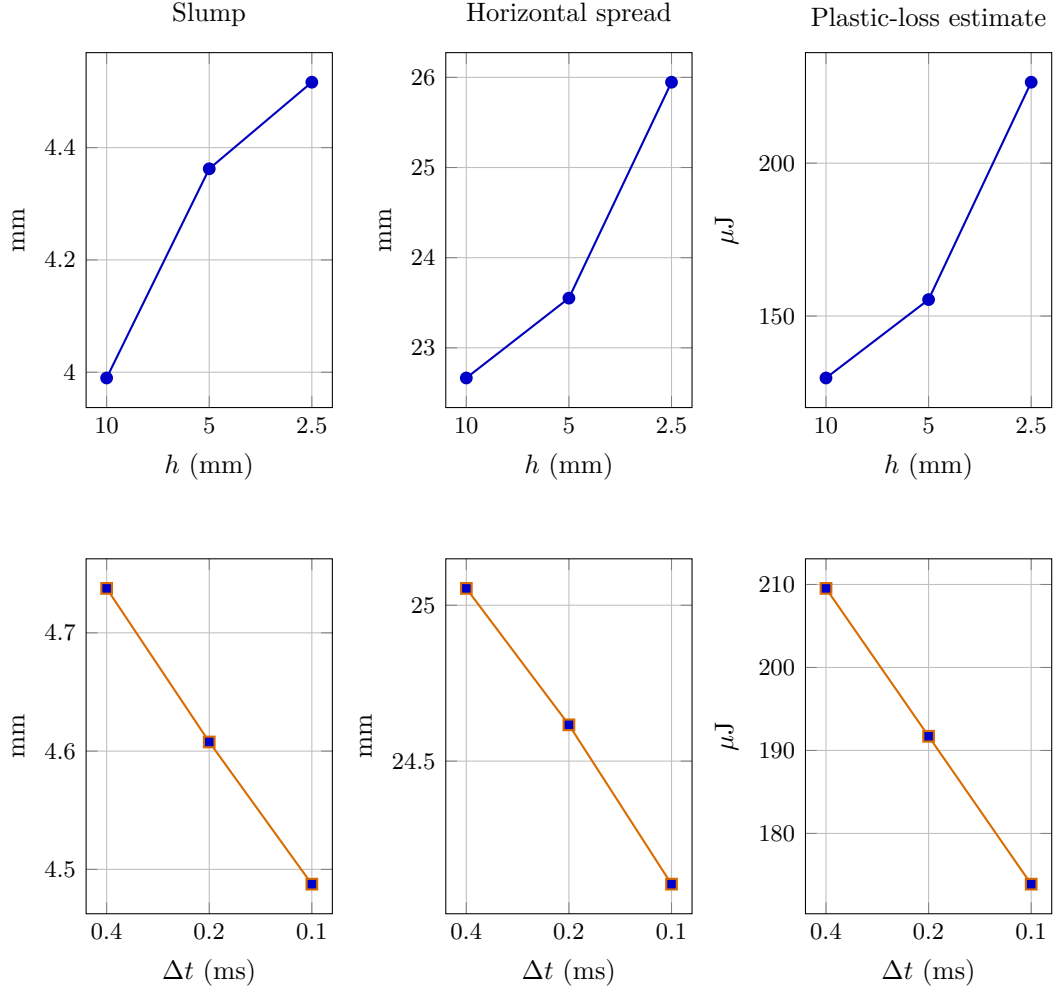


Figure 4: Frozen simulation evidence, *not* illustrative curves. Top row: spatial refinement at fixed $\Delta t = 0.05$ ms. Bottom row: temporal refinement at fixed $h = 5$ mm. Each axis becomes finer to the right; lines connect only the three observed values and are not fitted convergence curves. Vertical axes are independent between panels. The study remains unresolved because spread and plastic loss fail the specified final-two tests.

Final-two change	Slump	Spread	Plastic loss
Spatial	3.41%	9.23%	31.39%
Temporal	2.68%	2.12%	10.25%

Table 2: Final-two relative changes computed with (7.4). Required: each listed value strictly below 5%. Near-zero scales are 0.5 mm for lengths and 1 μ J for plastic loss.

At $h = 5$ mm and $\Delta t = 0.1$ ms, the recorded values include

$$E_N = -109.379 \text{ tJ}, \quad D_F = 35.486 \text{ tJ}, \quad (7.6)$$

$$D_P = 173.886 \text{ tJ}, \quad r_E = 84.476 \text{ tJ}. \quad (7.7)$$

The nonzero diagnostic is retained. It is neither a pass of energy closure nor an external-wall-work measurement. The six-case study took approximately 11.86 s on the producing arm64, 10-core, 16 GiB machine, measured through report evaluation (not the final report write). This is a tiny-fixture timing, not evidence for full-box or multi-day feasibility.

Implementation mapping and evidence identity. `core/src/mpm/fixtures.rs`: `FixtureSpec::shape_observables`, `slump_observables`, `hydrostatic_observables`. `python/litter_physics/verification.py`: `plan_cases`, `evaluate_observable`, `evaluate_axis`, `run_study`. The verbatim archived report is `docs/formal/data/refinement-report.json`; its source commit, clean-tree flag, native/Python hashes, dependency versions, hardware, all case values and unresolved verdicts are retained. The manuscript build verifies the archive’s SHA-256 before extracting any plot or table data.

8 Calibration, reduced responses, and uncertainty

8.1 What can currently be fitted

The Python calibration module contains bounded empirical fits including

$$U(t) = C[1 - e^{-k_u t}], \quad B(t) = 1 - e^{-k_b t}, \quad \tau(\dot{\gamma}) = \tau_y + K_s \dot{\gamma}^n. \quad (8.1)$$

Tilt observations supply a friction prior based on $\tan \theta$; they do not uniquely identify all multisphere sliding, rolling and shape parameters. Visit observations fit stage statistics and can be held out for personalization scoring. Synthetic observations remain explicitly synthetic; fit recovery from synthetic data cannot identify a real cat or material.

There is an important model discrepancy: the fitted breakup fraction $B(t)$ is exponential, whereas the household runtime uses the accumulated, whole-pellet threshold in (3.2). Transferring k_b between them does not automatically calibrate the distribution of actual breakup times. A simulator-based likelihood, an explicit heterogeneous threshold population, or measured justification of the reduction is still needed. Likewise, fitting a flow curve alone does not identify bulk elasticity, adhesion or porosity.

A prospective joint inverse problem can be written

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \sum_j \frac{[\mathcal{H}_j(\mathcal{S}(\theta)) - y_j]^2}{s_j^2} + \mathcal{R}(\theta), \quad (8.2)$$

where $\mathcal{S}$ is the simulator, $\mathcal{H}_j$ the measurement operator, y_j data, s_j uncertainty/repeatability, and $\mathcal{R}$ a justified prior. **Equation (8.2) is a proposed formulation, not an implemented end-to-end inference engine.** Current functions fit the separate empirical curves; unidentifiable parameter combinations must not be hidden by a low training residual.

8.2 Intended research-to-household response interface

The implemented standalone response-table class provides bounded multilinear interpolation on a full Cartesian grid. For a cell with lower/upper axis values x_j^-, x_j^+ , set $\lambda_j = (x_j - x_j^-)/(x_j^+ - x_j^-)$:

$$\hat{R}(\mathbf{x}) = \sum_{\mathbf{b} \in \{0,1\}^d} R_{\mathbf{b}} \prod_{j=1}^d \lambda_j^{b_j} (1 - \lambda_j)^{1-b_j}. \quad (8.3)$$

Inside the cell the weights are nonnegative and sum to one. For example, a hypothetical two-axis cell at $(\lambda_1, \lambda_2) = (1/4, 1/2)$ has weights $(3/8, 1/8, 3/8, 1/8)$ under a compatible corner ordering. Interpolation is exact for multilinear data, not for arbitrary nonlinear coupled material behavior.

The class validates grid completeness, units/physical bounds and evidence provenance. It refuses absent/synthetic evidence and extrapolation by default; explicit exploratory extrapolation is bounded to 10% of each axis span and still subject to response bounds. This mechanism does *not* produce validated evidence. No physical response tables are shipped, and neither native mode currently consumes these tables. The dashed bridge in Figure 1 is intentionally not an implemented feedback loop.

A defensible future acceptance chain is: constitutive/balance tests; independent space, timestep and domain-size studies; clean-surrogate measurements with held-out errors no greater than 20% or measured repeatability; then evidence-qualified response reduction and measured household tests. Material variation, parameter uncertainty, numerical error and behavior variability must be reported separately. A confidence interval on a fitted parameter does not absorb unknown model discrepancy.

Implementation mapping. `python/litter_physics/calibration.py`: `fit_uptake`, `fit_br`, `fit_rheology`, `friction_prior`, `score_personalization`, `apply_bundle`. `python/litter_physics/response_tables.py`: `ResponseTable`, `Axis`, `Evidence`, `Evaluation`.

9 Computational contract, reproducibility, and acceptance gaps

9.1 Execution and restart semantics

Python validates requests and delegates bulk JSON calls to the PyO3 native extension. Rust returns status, metrics, frames and full checkpoint state. Python publishes immutable run segments containing JSON, Parquet and Rerun artifacts, with configuration, code/build and dependency identity. Replay is served on loopback; its browser URL explicitly names the recording source. A rendered trajectory is an inspection tool, not a validation measurement.

Resuming requires matching frozen request/provenance and compatible native and Python builds. Native restore validates physical/history/adaptive state as well as geometry and baseline ledgers. A checkpoint lacking required history is rejected instead of assigning unearned zero dissipation or limiter counts. Atomic exclusive publication prevents competing writers from silently replacing an existing run.

Research jobs have a one-hour local per-job target budget. Checks are cooperative and include initialization/output budgeting, but do not preempt an expensive operation or guarantee an exact wall-clock cutoff during finalization. An incomplete run preserves a checkpoint and reports incompleteness. It must not quietly switch to cheaper constitutive laws. Resource caps are guards, not performance guarantees; a schema-valid request may still be infeasible.

9.2 Scaling calculation, not an extrapolated benchmark

For the 20 mm cube with two points per cell per axis,

$$N_p = \prod_{\alpha=1}^3 \frac{L_\alpha}{h/2} = \left(\frac{2L}{h}\right)^3. \quad (9.1)$$

This gives 64, 512 and 4096 points at $h = 10, 5, 2.5$ mm. A P2G/G2P sweep touches 27 stencil nodes per point. Ignoring constants, occupied-grid work and output,

$$\text{transfer work} \sim N_p \frac{T}{\Delta t}; \quad \begin{cases} O(h^{-4}), & \Delta t \propto h \text{ under an acoustic limit,} \\ O(h^{-5}), & \Delta t \propto h^2 \text{ under a viscous limit.} \end{cases} \quad (9.2)$$

Therefore halving h can multiply this work by 16 or 32, not merely 8. Neither scaling law is a measured runtime prediction: active sparsity, cache behavior, return mapping, retries, sorting, contact and output all matter. The present household scheduler does not establish adaptive multi-day idle integration or actual-size runtime acceptance.

9.3 Explicitly unresolved physics and numerical acceptance

- Hydrostatic bottom-boundary accuracy and liquid per-particle pressure noise remain unresolved. This document introduces no smoothing or weaker acceptance metric to make them disappear.
- Slump spatial spread and plastic-loss refinement fail the recorded threshold. Domain-size and combined space-time studies are still missing.
- The energy ledger is a mixed numerical diagnostic, not physical closure. Coupled work, leakage and contact/transfer energy require further analysis.
- Porous fines, capillary transport, absorption, wet fragmentation and adhesion are not research continuum phases. Research fines are explicitly unsupported; multiple research pellets are not implemented.
- Household paste stresses, resolved spreading/reaction, calibrated slot flux, measured settled packing, entrances/enclosures/pads and quantitative burial/exposure metrics remain incomplete.
- Per-cat motion/force identification, measured clean-surrogate validation, held-out household predictions and a validated response bridge are absent.

These gaps delimit what can be inferred from the formal equations. Conservation, a successful compile, a smooth replay or a small synthetic fit residual cannot replace the missing acceptance evidence.

A Equation-to-code navigation

Links below pin the scientific source baseline, so later implementation changes do not silently alter the meaning of the derivations. More detailed symbol mappings appear next to the corresponding sections.

Formal component	Source and role
State map (1.1), inputs	python/litter_physics/models.py : requests and domain validation. core/src/lib.rs : native mode dispatch.
Rigid geometry and contacts, (2.1)–(2.11)	core/src/dem.rs : multisphere geometry, forces, history, integration and slots.
Uptake/damage/swelling, (3.1)–(3.3)	core/src/household.rs : stateful boxes, events and moisture histories.
Reduced transport, (3.4)	core/src/household/field.rs : free water, fines, occupancy and drawer flux.
Paste and liquid constitutive laws	core/src/mpm/constitutive.rs : material conversion, local return and EOS.
Stencil and transfers, (5.3)–(5.10)	core/src/mpm/grid.rs , core/src/mpm/particles.rs , core/src/mpm/solver.rs : grid support, particle history and stepping.
Rigid impulse exchange, (6.2)	core/src/mpm/rigid.rs : research pellet, wall contact and grid coupling.
Research residuals, (6.5)–(6.7)	core/src/mpm/solver.rs : signed energy channels and state validation. core/src/research.rs : closed-wire-fixture checks and checkpoint restore.
Fixtures and observables, (7.1)	core/src/mpm/fixtures.rs : initialization, surface proxies and diagnostic reference fields.
Refinement criterion, (7.4)	python/litter_physics/verification.py : case plans, fail-closed evidence and verdicts.
Behavior and empirical calibration	python/litter_physics/behavior.py , python/litter_physics/calibration.py : visit distributions, fits and evidence labels.
Response interpolation, (8.3)	python/litter_physics/response_tables.py : standalone evidence-bounded interpolation, not native dynamics.
Run artifacts and replay	python/litter_physics/runner.py , python/litter_physics/artifacts.py , python/litter_physics/provenance.py , python/litter_physics/viewer.py : orchestration and inspection.

B Rebuilding the document and reproducing the experiment

From the repository root, with the locked Python environment and Tectonic installed:

```
make manuscript
uv run python docs/formal/build.py --check
uv run litter-physics verify examples/verification_slump.yaml \
    --out outputs/reproduce-formal-slump
```

The build checks the archived report, generates deterministic CSV/LaTeX data, and compiles this PDF. Method diagrams are vector TikZ drawings; the kernel plot is analytic; Figure 4 uses the archived native results. No new physics run is hidden inside document compilation. The baseline study’s expected verification exit status is 1 (complete but unresolved), not 0. Fresh run directories must be empty. Exact wall times and PDF bytes need not match across platforms or tool versions; numerical data and provenance remain inspectable.

The report’s SHA-256 is

```
aa6030d7ea1af0919343d3c4732855bce3eccc3c69310a7bab116dc0015bce8.
```

`docs/formal/README.md` documents compiler requirements, generated artifacts, checks and the intentional historical source pin.

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